MOHAMMAD ROBATI SHIRZAD

SOFTWARE ENGINEERING II

DR. TAROMIRAD

MINING SOFTWARE REPOSITORIES

- Prior experiences and dominant patterns are the driving force for many decision-processes in modern software organizations.
- Practitioners often rely on their experience, intuition and gut feeling in making important decisions.
- Managers allocate development and testing resources based on their experience in previous projects and their intuition about the complexity of the new project relative to prior projects.
- Developers commonly use their experience when adding a new feature or fixing a bug.
- Testers usually prioritize the testing of features that are known to be error prone based on field and bug reports.

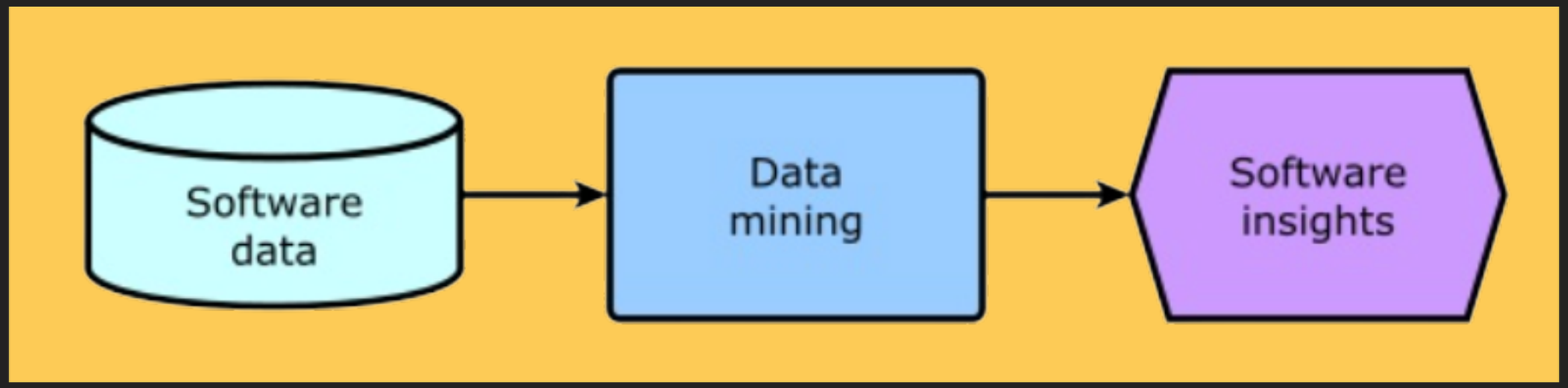
- Software repositories contain a wealth of valuable information about software projects.
- Using the information stored in these repositories, practitioners can depend less on their intuition and experience and depend more on historical and field data.

SOFTWARE REPOSITORIES

- Historical repositories
- Run-time repositories
- Code repositories

MINING SOFTWARE REPOSITORIES

- The Mining Software Repositories (MSR) field analyzes and cross-links the rich data available in software repositories to uncover interesting and actionable information about software systems and projects.

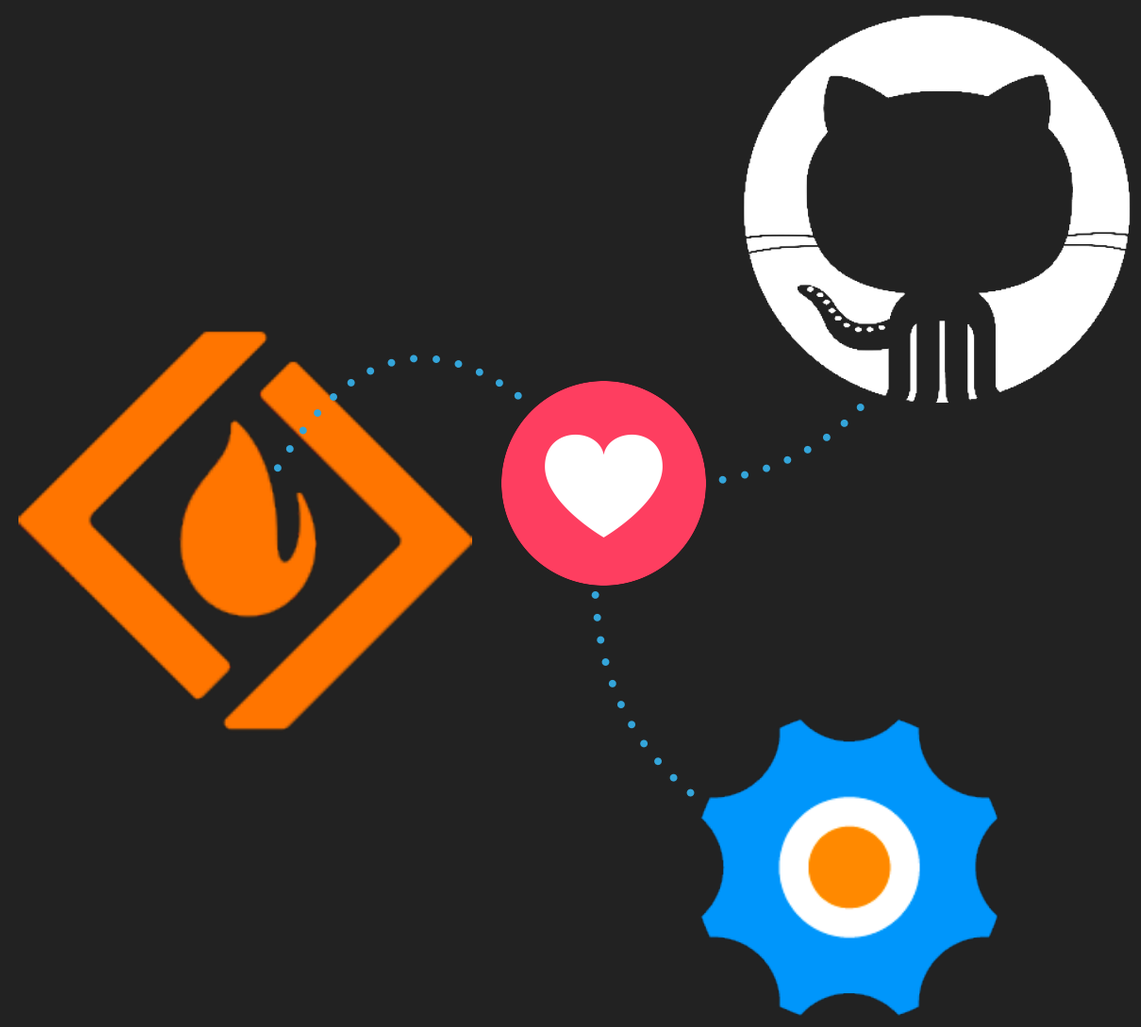


MINING SOFTWARE REPOSITORIES

- MSR researchers aim to transform these repositories from static record-keeping ones into active repositories which can guide decision processes in modern software projects.
- For example, historical repositories are used to track the history of a bug or a feature, but are not commonly used to determine the expected resolution time of an open bug based on the resolution time of previously-closed bugs.
- The MSR field is rapidly taking a central and important role in supporting software development practice and soft- ware engineering research.

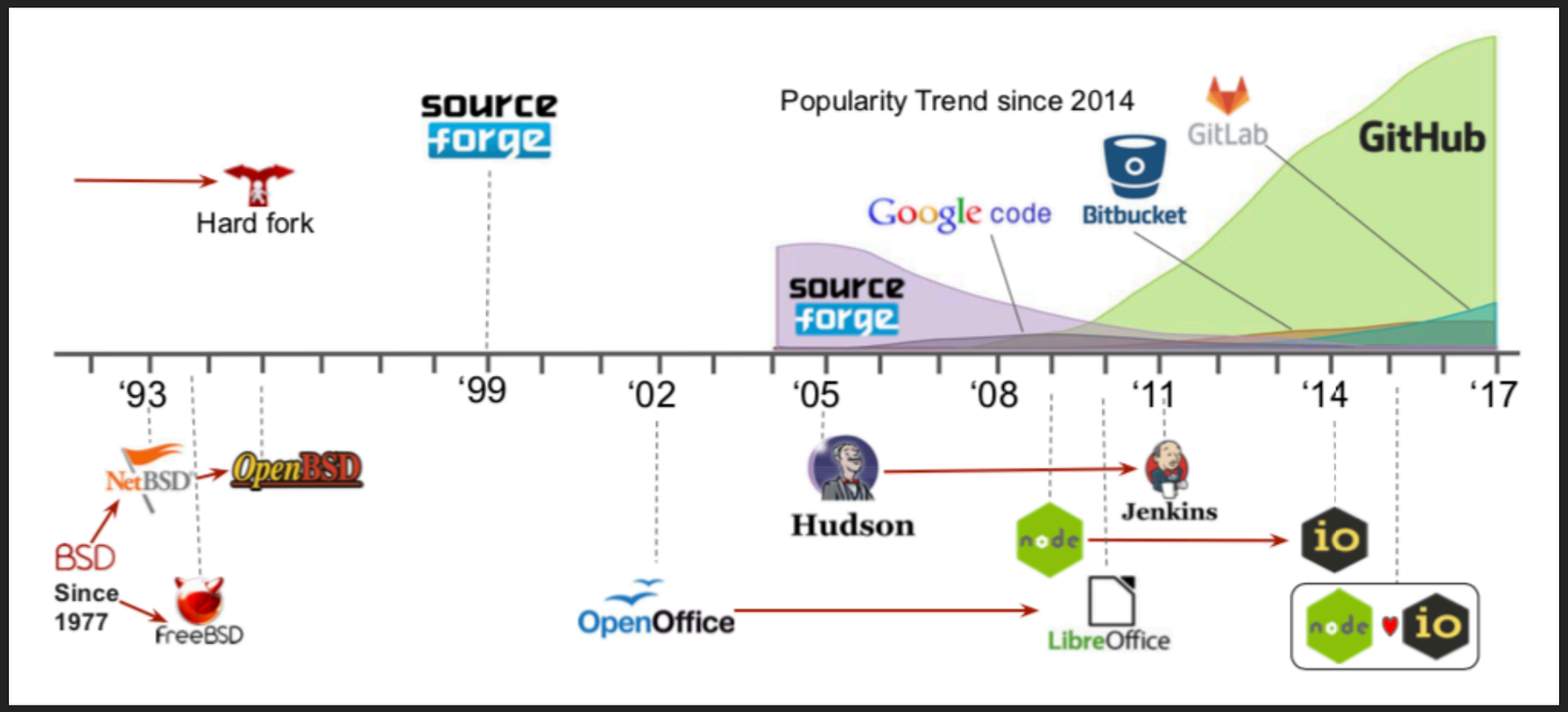
WHERE IT STARTED ?

- The Mining Software Repositories (MSR) field is maturing thanks to the rich, extensive, and readily available software repositories.
- Problems though:
- 1. Enterprise limit
- 2. Hard!



HOW HAS FORKING CHANGED IN THE LAST 20 YEARS?

- Hard forks / Social forks
- Hard fork : fragment a community ?
- Hard fork : survival-of-the-fittest model ?



DO TIME OF DAY AND DEVELOPER EXPERIENCE AFFECT COMMIT BUGGINESS?

- Commit metadata is a rich source of social characteristics, including the commit's time of day and the experience and commit frequency of its author.
- Midnight – 4AM (double-check their own late- night commits)
- 7AM – Noon
- Daily committing produces less buggy code than day-job developing.

WHAT CAN OSS MAILING LISTS TELL US?

- Used LIWC on Apache HTTP Server mailing lists
- The two developers that were responsible for two major Apache releases had similar personalities.
- New, Current, and Departing Developers.
- The 1.3 release had higher pre-release levels of optimism

WHERE IS IT GOING ?

- Taming the Complexity of MSR
- Going Beyond Code and Bugs
- Showing the Value of Repositories
- Easing the Adoption of MSR

IN CONCLUSION...

- Software repositories have traditionally been used for archival purposes.
- The MSR field has shown that these repositories could be mined to uncover useful patterns and actionable information about software systems and projects.
- MSR researchers have proposed techniques which augment traditional software engineering data, techniques and tools, in order to solve important and challenging problems, such as identifying bugs, and reusing code, which practitioners must face and solve on a daily basis.

REFERENCES

- The road ahead for Mining Software Repositories, Ahmed E. Hassan, 2008 Frontiers of Software Maintenance
- Do time of day and developer experience affect commit bugginess? J Eyolfson, L Tan, P Lam, 2011
- How has forking changed in the last 20 years? a study of hard forks on GitHub, S Zhou, B Vasilescu, C Kästner, 2020
- What can OSS mailing lists tell us? A preliminary psychometric text analysis of the Apache developer mailing list, PC Rigby, AE Hassan, 2007

BOA PROGRAMMING LANGUAGE

- A programming language for MSR

boa

Run an Example

What are the ten most used programming languages?

Boa Source Code

```
# Counting the 10 most used programming languages
p: Project = input;
counts: output top(10) of string weight int;

foreach (i: int; def(p.programming_languages[i]))
    counts << p.programming_languages[i] weight 1;
```

Input Dataset (use the SMALL dataset when testing queries!)

2015 September/GitHub

Run Program NOTE: All data submitted to this site is subject to our privacy policy.

```
counts = JavaScript, 1473096.0
counts = Ruby, 889738.0
counts = Shell, 700831.0
counts = Python, 620213.0
counts = Java, 554864.0
counts = PHP, 489082.0
counts = C, 419044.0
counts = CSS, 354715.0
counts = C++, 333877.0
counts = Perl, 274174.0
```

THANK YOU